Online Coupling of ML Cloud Microphysics in a Climate Experiment

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1 Introduction and objectives

Climate models are beginning to explicitly resolve kilometre-scale storms and clouds, however, cloud microphysical processes still have to be parameterised. Their representation strongly influences both precipitation and radiation and remains an important source of uncertainty in modern climate models (Naumann, 2025). In the ICON model, cloud microphysics are represented by bulk moment schemes that simplify droplet interactions (Seifert, 2006).

Physically consistent droplet level representations remain computationally infeasible for climate simulations. SuperdropNet is an ML emulator for the warm rain collisions, trained on superdroplet simulations (Sharma, 2025). SuperdropNet coupled to ICON has been evaluated in a cloud microphysics warm-bubble test case (Arnold, 2025).

In this work, we perform a realistic climate experiment with SuperdropNet coupled online to ICON at 10 km resolution and discuss changes in precipitation, cloud properties, and radiation. We investigate the generalisation capabilities of the ML emulator and the stability of the coupled model. The results will inform the future development of hybrid climate models, in which subgrid-processes are replaced with ML emulators.

2 Methods and data

We create a hybrid model, ICON-SuperdropNet, by coupling SuperdropNet (Sharma, 2025) with ICON (Release 2024.10). In the cloud microphysics module, warm-rain collisions are replaced by SuperdropNet calls via the FTorch library (Atkinson, 2025). SuperdropNet is constrained to enforce the mass conservation of the total water content.

We perform an AMIP-style experiment with prescribed sea surface temperature (SST), sea ice, and greenhouse gases, consisting of a 14-day spin-up followed by 7 days of simulation using both ICON-SuperdropNet and the two-moment bulk scheme for reference.

3 Preliminary results and discussion

The ICON-SuperdropNet simulation is stable, and can handle conditions beyond the range of the ML emulator's training data. The surface precipitation pattern and the zonal mean are comparable to the reference simulation, highlighting physical consistency.

Figure 1a shows the frequency spectrum of precipitation for both simulations, and compares it with IMERG observations. Both simulations overestimate large precipitation events, but ICON-SuperdropNet shifts the spectrum slightly towards the observations.

ICON-SuperdropNet produces more cloud water mass (about twice as much in the tropics) than the reference simulation (Figure 1b), yet the amount of cloud ice is similar. We hypothesise that this is related to the faster cloud-to-rain transition observed in the offline evaluation of SuperdropNet. The cloud water mass increase leads to substantially more shortwave radiation being reflected in the tropics.

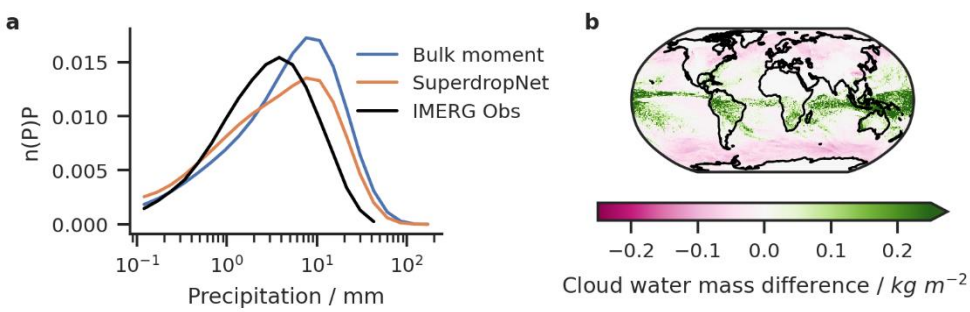

Figure 1. Comparing ICON-SuperdropNet and ICON-BulkMoment in an AMIP-style experiment.

We will further investigate the impact on lapse rate and droplet distribution. Finally, we plan to extend these experiments to a warmer world by increasing SST by 4K.

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